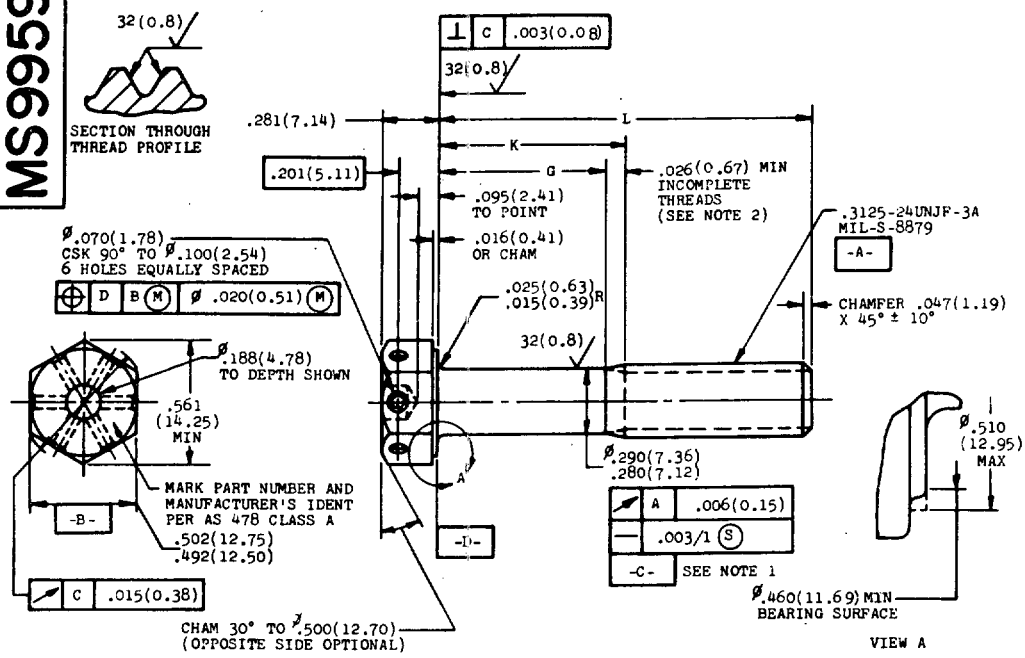
MS9959

FED. SUP CLASS 5306



1. FOR PART NUMBERS MS9959-04 THRU MS9959-19 THE THREAD PD SHALL REPLACE DATUM C.
2. INCOMPLETE THREADS NOT TO ENTER FILLET.
3. MATERIAL: STEEL AMS 6322.
4. HARDNESS: ROCKWELL C26-32.
5. FINISH: CADMIUM PLATE AMS 2400. DIMENSIONS SPECIFIED ARE AFTER PLATING.
6. MANUFACTURING SPECIFICATION: AMS 7452 EXCEPT HEAD SHALL BE UPSET.
7. HEAD TO SHANK FILLET SHALL BE COLD WORKED.
8. MAGNETIC PARTICLE INSPECTION PER AMS 2640 AFTER PLATING.
9. SURFACE TEXTURE: ANSI B46.1-1962. UNLESS OTHERWISE SPECIFIED, SURFACES TO BE 125 MICROINCHES (3.2 MICROMETERS) EXCEPT UPSET HEAD.
10. BREAK SHARP EDGES .003-.015 (0.08-0.38) UNLESS OTHERWISE SPECIFIED.
11. DIMENSIONS IN INCHES. METRIC CONVERSIONS ARE IN PARENTHESES. UNLESS OTHERWISE SPECIFIED: TOLERANCES; LINEAR DIMENSIONS ±.010 (0.25), ANGULAR DIMENSIONS ±5° INTERNATIONAL SYSTEM UNITS (SI) SHOWN ARE FOR REFERENCE ONLY.
12. DIMENSIONING AND TOLERANCING: ANSI Y14.5-1966, Ø = DIAMETER.
13. DO NOT USE UNASSIGNED PART NUMBERS.

AS AND AMS ARE SOCIETY OF AUTOMOTIVE ENGINEERS, INC., PUBLICATIONS.
THIS STANDARD WAS DEVELOPED COOPERATIVELY WITH THE MILITARY SERVICES BY THE SAE AEROSPACE PART STANDARDS DIVISION.

User activities: Army, Navy, Air Force, DSA

Review activities: Army, Navy, Air Force, DSA

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

APPROVED 27 Jun 74 REVISED

P.A. Air Force – 11 Other Cust Army – AV Navy – AS	TITLE BOLT, MACHINE, HEXAGON HEAD, DRILLED, 6 HOLE, PD SHANK, STEEL AMS 6322, CADMIUM PLATED, .3125-24UNJF-3A	MILITARY STANDARD MS9959
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 1 OF 2

DD FORM 672-1 ASD use only

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

5306-0494

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:28Z
Check the source to verify that this is the current version before use.

MS9959

FED. SUP CLASS 5306

User activities: Army, Navy, Air Force, DSA

Review activities: Army, Navy, Air Force, DSA

PART NUMBER	L		G MIN		K MAX		APPROX MASS		PART NUMBER	L		G MIN		K MAX		APPROX MASS	
	IN.	(mm)	IN.	(mm)	IN.	(mm)	LB/100	kg/100		IN.	(mm)	IN.	(mm)	IN.	(mm)	LB/100	kg/100
MS9959-04	.500	12.70	.025	0.64	.108	2.74	2.36	1.070	MS9959-31	2.375	60.32	1.957	34.47	1.500	38.10	5.75	2.608
MS9959-05	.562	14.27	.025	0.64	.108	2.74	2.48	1.125	MS9959-32	2.500	63.50	1.482	37.65	1.625	41.27	5.97	2.708
									MS9959-33	2.625	66.68	1.607	40.82	1.750	44.45	6.20	2.812
MS9959-06	.625	15.88	.025	0.64	.108	2.74	2.59	1.175	MS9959-34	2.750	69.85	1.732	44.00	1.875	47.62	6.42	2.912
MS9959-07	.688	17.48	.025	0.64	.108	2.74	2.70	1.225	MS9959-35	2.875	73.02	1.857	47.17	2.000	50.80	6.62	3.003
MS9959-08	.750	19.05	.025	0.64	.108	2.74	2.81	1.275									
MS9959-09	.812	20.62	.025	0.64	.108	2.74	2.93	1.329	MS9959-36	3.000	76.20	1.982	50.35	2.125	53.97	6.88	3.121
MS9959-10	.875	22.22	.025	0.64	.108	2.74	3.04	1.379	MS9959-37	3.125	79.38	2.107	53.52	2.250	57.15	7.10	3.220
									MS9959-38	3.250	82.55	2.232	56.70	2.375	60.32	7.33	3.325
MS9959-11	.938	22.82	.025	0.64	.108	2.74	3.15	1.429	MS9959-39	3.375	85.72	2.357	59.87	2.500	63.50	7.55	3.425
MS9959-12	1.000	25.40	.025	0.64	.125	3.17	3.27	1.483	MS9959-40	3.500	88.90	2.482	63.05	2.625	66.67	7.78	3.529
MS9959-13	1.062	26.97	.044	1.12	.188	4.77	3.38	1.533									
MS9959-14	1.125	28.58	.107	2.72	.250	6.35	3.49	1.583	MS9959-41	3.625	92.08	2.607	66.22	2.750	69.85	8.01	3.633
MS9959-15	1.188	30.18	.169	4.30	.312	7.92	3.60	1.633	MS9959-42	3.750	95.25	2.732	69.40	2.875	73.02	8.23	3.733
									MS9959-43	3.875	98.42	2.857	72.57	3.000	76.20	8.46	3.837
MS9959-16	1.250	31.75	.232	5.90	.375	9.52	3.72	1.687	MS9959-44	4.000	101.60	2.982	75.75	3.125	79.37	8.68	3.937
MS9959-17	1.312	33.32	.295	7.50	.438	11.12	3.83	1.737	MS9959-45	4.125	104.78	3.107	78.92	3.250	82.55	8.91	4.042
MS9959-18	1.375	34.92	.357	9.07	.500	12.70	3.94	1.787									
MS9959-19	1.438	36.52	.419	10.65	.562	14.27	4.06	1.842	MS9959-46	4.250	107.95	3.232	82.10	3.375	85.72	9.13	4.141
MS9959-20	1.500	38.10	.482	12.25	.625	15.88	4.17	1.891	MS9959-47	4.375	111.12	3.357	85.27	3.500	88.90	9.36	4.246
									MS9959-48	4.500	114.30	3.482	88.45	3.625	92.07	9.59	4.350
MS9959-21	1.562	39.67	.545	13.85	.688	17.47	4.28	1.941	MS9959-49	4.625	117.48	3.607	91.62	3.750	95.25	9.81	4.450
MS9959-22	1.625	41.28	.607	15.42	.750	19.05	4.39	1.991	MS9959-50	4.750	120.65	3.732	94.80	3.875	98.42	10.04	4.554
MS9959-23	1.688	42.88	.669	17.00	.812	20.62	4.51	2.046									
MS9959-24	1.750	44.45	.732	18.60	.875	22.22	4.61	2.097	MS9959-51	4.875	123.83	3.857	97.97	4.000	101.60	10.26	4.654
MS9959-25	1.812	46.05	.795	20.20	.938	23.82	4.72	2.145	MS9959-52	5.000	127.00	3.982	101.15	4.125	104.77	10.49	4.758
									MS9959-53	5.125	130.18	4.107	104.32	4.250	107.95	10.71	4.858
MS9959-26	1.875	47.62	.857	21.77	1.000	25.40	4.83	2.200	MS9959-54	5.250	133.35	4.232	107.50	4.375	111.12	10.94	4.962
MS9959-27	1.938	49.22	.919	23.35	1.062	26.97	4.94	2.250	MS9959-55	5.375	136.52	4.357	110.67	4.500	114.30	11.17	5.067
MS9959-28	2.000	50.80	.982	24.95	1.125	28.57	5.05	2.300									
MS9959-29	2.125	53.98	1.107	28.12	1.250	31.75	5.27	2.404	MS9959-56	5.500	139.70	4.482	113.85	4.625	117.47	11.39	5.186
MS9959-30	2.250	57.15	1.232	31.30	1.375	34.92	5.50	2.504	MS9959-57	5.625	142.88	4.607	117.02	4.750	120.65	11.62	5.271
									MS9959-58	5.750	146.05	4.732	120.20	4.875	123.82	11.84	5.371
									MS9959-59	5.875	149.22	4.857	123.37	5.000	127.00	12.07	5.475
									MS9959-60	6.000	152.40	4.982	126.55	5.125	130.17	12.29	5.575

AS AND AMS ARE SOCIETY OF AUTOMOTIVE ENGINEERS, INC., PUBLICATIONS.
THIS STANDARD WAS DEVELOPED COOPERATIVELY WITH THE MILITARY SERVICES BY THE SAE AEROSPACE PART STANDARDS DIVISION.

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

APPROVED 27 Jun 74 REVISED

P.A. Air Force - 11

Other Cust
Army - AV
Navy - AS

PROCUREMENT SPECIFICATION

TITLE

BOLT, MACHINE, HEXAGON HEAD, DRILLED, 6 HOLE, PS SHANK, STEEL AMS 6322, CADMIUM PLATED, .3125-24UNJF-3A

SUPERSEDES:

MILITARY STANDARD

MS9959

SHEET 2 OF 2

DD FORM 672-1

ASD use only

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

5306-0494

☆U.S. GOVERNMENT PRINTING OFFICE: 1981 — 703-023/7749

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:28Z
Check the source to verify that this is the current version before use.